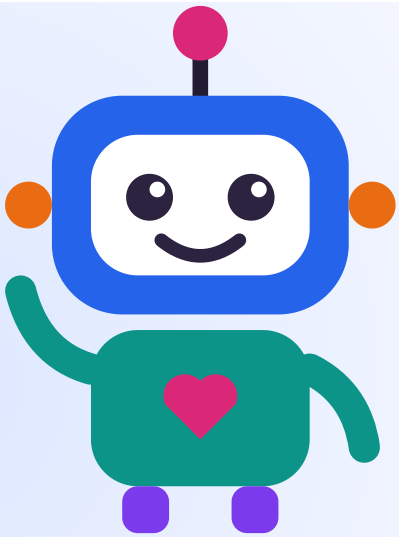


A Year of Robots

My robot activity book

This book belongs to _____

50 weeks · build, play, talk, level up · https://pandagardenio.github.io/robokids/web/

🧭 How this book works

- **One build session per week** (30–45 min), plus one or two 10–15 minute play sessions.
- **Every week has the same shape:** Build → Play → Talk about it → Level up → Robot Journal.
- **Kid leads, parent assists.** They choose names, colors, rules and numbers. You handle batteries, wiring and loading the programs.
- **Weeks 10, 20, 30, 40 and 50 are showcases** with slack built in. Skip or repeat any week.
- **Track progress** with the progress checklist.

- **Every week has two pages:** the activity (for doing it together) and a Robot Journal page (for your kid to draw and write).
- **Programs** (from week 11) are loaded on a laptop from the website. A grown-up does that part; the week's page says where.
- **Tick the box** at the top of each week when it's done, and color a circle on the fridge poster.

> 🦺 **Safety first.** Coin cells (CR2032) are dangerous if swallowed: grown-ups hold them, tape them in and store them out of reach. Adults handle craft knives, hot glue and wire stripping.

🗓️ The year

💡 **Phase 1 · Circuits and Motors**	Weeks 1–10
📟 **Phase 2 · micro:bit First Steps**	Weeks 11–20
🚗 **Phase 3 · The Robot Car**	Weeks 21–30
🦾 **Phase 4 · Smarter Robots**	Weeks 31–40
🧠 **Phase 5 · Robots That Learn**	Weeks 41–50
🐭 **Bonus · Robot Mouse**	after week 37



PHASE 1 · WEEKS 1–10

Circuits and Motors

Electricity flows in a loop and makes things glow and move. You handle batteries and wiring; your kid connects, tests and decorates.

PHASE 1 · CIRCUITS AND MOTORS

1 LED Robot Eyes

~45 min build + short play sessions during the week · Skill: A circuit is a loop

Done! ☐

🧰 You need

- ☐ 2 LEDs (5 mm, any color)
- ☐ 2 CR2032 coin cells
- ☐ Masking tape
- ☐ A cardboard robot head (any box)

🔧 Build together

TOGETHER

1. Slide the LED legs around the coin cell: long leg on the + side (the flat, printed side).
2. It lights? Wrap it tightly in tape so the battery cannot fall out.
3. Poke two holes in the box and push the lit LEDs through as eyes.

🎮 Play

KID TIME

Turn off the room lights and let the robot 'look around' the house. Flip one LED around: why does it stay dark?

💬 Talk about it

TOGETHER

What has to touch what to make the light turn on?

⭐ Level up

KID TIME

Try different colors. Does a red LED and a blue LED look different in the dark?

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
LED	A tiny light. It only works one way round: the long leg goes to plus (+).	A tiny light that only works the right way round.
Circuit	A complete loop that electricity travels around: out of the battery, through the parts, and back in. Any gap stops everything.	A round trip for electricity. If the path is broken, nothing works.
Coin cell (CR2032)	A flat, round battery the size of a big button. Dangerous if swallowed: grown-ups handle it (see Safety).	A flat battery. Only grown-ups hold it.
Plus (+) and minus (–)	The two ends of a battery. Electricity leaves from one end and has to find its way back to the other.	The battery's way out and way back in.

Week 1 · LED Robot Eyes

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

PHASE 1 · CIRCUITS AND MOTORS

2 Paper Circuit Card

~45 min build + short play sessions during the week · Skill: Circuits on paper

Done! ☐

🧰 You need

- ☐ Copper tape (5 mm, conductive adhesive)
- ☐ 1 LED
- ☐ 1 CR2032 coin cell
- ☐ Card stock
- ☐ Binder clip

🔧 Build together

TOGETHER

1. Draw a robot on the card. Mark where the eye (LED) goes.
2. Run two copper tape lines from the eye to a corner (don't let them touch).
3. Bend the LED legs flat and tape them onto the lines; long leg on the + line.
4. Put the coin cell in the corner, + side up, and fold the corner over it. Clip it.

🎮 Play

KID TIME

Give the glowing card to someone in the family. Can they figure out how it works?

💬 Talk about it

TOGETHER

What happens if the two copper lines touch each other?

⭐ Level up

KID TIME

Add a second LED eye on the same two lines.

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
LED	A tiny light. It only works one way round: the long leg goes to plus (+).	A tiny light that only works the right way round.
Coin cell (CR2032)	A flat, round battery the size of a big button. Dangerous if swallowed: grown-ups handle it (see Safety).	A flat battery. Only grown-ups hold it.
Copper tape	Sticky metal tape that electricity can travel along, like a flat wire.	Sticky tape that electricity can walk on.
Plus (+) and minus (–)	The two ends of a battery. Electricity leaves from one end and has to find its way back to the other.	The battery's way out and way back in.

Week 2 · Paper Circuit Card

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

PHASE 1 · CIRCUITS AND MOTORS

3 The Robot Switch

~45 min build + short play sessions during the week · Skill: Open and closed circuits

Done! ☐

🧰 You need

- ☐ Last week's card
- ☐ Extra copper tape
- ☐ Paper fastener or pushbutton (optional)

🔧 Build together

TOGETHER

1. Cut one copper line so there is a gap.
2. Make a paper flap with copper tape underneath that covers the gap when pressed.
3. Press the flap: light on. Let go: light off.

🎮 Play

KID TIME

'Wink game': you call out 'wink!' and they press. Then switch roles and try Morse-style blink patterns (short, long).

💬 Talk about it

TOGETHER

Where else at home do you see switches?

⭐ Level up

KID TIME

Make a switch that turns on when the card is opened, like a magic greeting card.

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
Switch	Something that opens or closes a gap in a circuit, to turn things off and on.	A little door for electricity: open means off, closed means on.
Circuit	A complete loop that electricity travels around: out of the battery, through the parts, and back in. Any gap stops everything.	A round trip for electricity. If the path is broken, nothing works.
Copper tape	Sticky metal tape that electricity can travel along, like a flat wire.	Sticky tape that electricity can walk on.

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

PHASE 1 · CIRCUITS AND MOTORS

4 Two Eyes, Two Ways

~45 min build + short play sessions during the week · Skill: Series vs parallel

Done! ☐

🧰 You need

- ☐ 4 LEDs
- ☐ 2 AA battery holder with switch + 2 AA batteries
- ☐ Alligator clip leads (6)

🔧 Build together

TOGETHER

1. Connect two LEDs one after the other (series). Note how bright.
2. Now connect them side by side (parallel). Note how bright.
3. Compare: which robot eyes are brighter?

🎮 Play

KID TIME

Make a 'sleepy robot' (series, dim) and an 'awake robot' (parallel, bright) and act them out.

💬 Talk about it

TOGETHER

Why did one way look dimmer?

⭐ Level up

KID TIME

Unscrew one LED in each version. What happens to the other one?

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
Series	Parts connected one after the other in a single line. They share the battery's push, so LEDs glow dimmer, and if one goes out they all do.	Holding hands in a line: everyone shares.
Parallel	Parts connected side by side, each with its own path to the battery. LEDs glow brighter, and one going out doesn't stop the others.	Everyone gets their own path.
LED	A tiny light. It only works one way round: the long leg goes to plus (+).	A tiny light that only works the right way round.
Switch	Something that opens or closes a gap in a circuit, to turn things off and on.	A little door for electricity: open means off, closed means on.
Alligator clip	A wire with a spring clip at each end, for connecting parts without tools.	A wire with crocodile teeth.

Week 4 · Two Eyes, Two Ways

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

PHASE 1 · CIRCUITS AND MOTORS

5 Bristlebot

~45 min build + short play sessions during the week · Skill: Vibration makes motion

Done! ☐

🧰 You need

- ☐ Toothbrush head (cut off the handle)
- ☐ Mini vibration motor (coin type)
- ☐ CR2032 coin cell
- ☐ Double-sided foam tape
- ☐ Googly eyes

🔧 Build together

TOGETHER

1. Stick the coin cell on top of the brush head with foam tape.
2. Stick the vibration motor on the front.
3. Touch one motor wire to the bottom of the battery, tape the other wire to the top to start it.

🎮 Play

KID TIME

Bristlebot races on a baking tray. Build a little maze with LEGO bricks or pencils.

💬 Talk about it

TOGETHER

The motor only shakes. How does the bot move forward?

⭐ Level up

KID TIME

Bend the bristles forward or backward. Does it change direction?

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
Vibration motor	A tiny motor with a lopsided weight that makes it shake, like a phone buzzing.	A motor that shakes instead of spins.
Coin cell (CR2032)	A flat, round battery the size of a big button. Dangerous if swallowed: grown-ups handle it (see Safety).	A flat battery. Only grown-ups hold it.

Week 5 · Bristlebot

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

PHASE 1 · CIRCUITS AND MOTORS

6 Scribble Bot

~45 min build + short play sessions during the week · Skill: Unbalanced motors

Done! ☐

🧰 You need

- ☐ Paper cup
- ☐ Small DC motor (3–6 V)
- ☐ AA battery holder + 2 AA
- ☐ 3 felt-tip markers
- ☐ Hot glue (parent)
- ☐ Cork or eraser piece

🔧 Build together

TOGETHER

1. Tape the markers around the cup as legs, caps off.
2. Glue the battery holder on top and the motor on the side.
3. Push a cork piece off-center onto the motor shaft so it wobbles.
4. Place on big paper and switch on.

🎮 Play

KID TIME

Make robot art. Frame the best one in the Robot Journal.

💬 Talk about it

TOGETHER

Why does putting the cork off-center make it shake?

⭐ Level up

KID TIME

Move the cork to the center. What changes in the drawing?

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
Off-center	Weight that isn't in the middle of something spinning, which makes it wobble and shake.	Lopsided, so it wobbles.
DC motor	A motor that spins when connected to a battery. Swap its two wires and it spins the other way.	A spinner that runs on batteries.
Volts (V)	How hard a battery pushes electricity. One AA battery is 1.5 V; two make 3 V.	How strong the battery's push is.
Shaft	The metal rod that sticks out of a motor and spins.	The motor's spinning nose.

Week 6 · Scribble Bot

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

PHASE 1 · CIRCUITS AND MOTORS

7 Wobble-Bot Sumo

~45 min build + short play sessions during the week · Skill: Weight and balance

Done! ☐

🧰 You need

- ☐ 2 scribble bots or bristlebots
- ☐ A shoebox lid as the ring
- ☐ Modeling clay

🔧 Build together

TOGETHER

1. Add clay to make each bot heavier or lower.
2. Draw a circle in the lid as the sumo ring.

🎮 Play

KID TIME

Sumo: both bots start in the ring, last one inside wins. Best of five.

💬 Talk about it

TOGETHER

Did the heavier robot win more?

⭐ Level up

KID TIME

Design a bumper that pushes better (cardboard plow, rubber band).

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
Balance	How weight is spread out. Low and heavy robots are harder to tip over.	Low and heavy is hard to push over.

Week 7 · Wobble-Bot Sumo

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

PHASE 1 · CIRCUITS AND MOTORS

8 Fan Car

~45 min build + short play sessions during the week · Skill: Push and measuring

Done! ☐

🧰 You need

- ☐ Cardboard chassis
- ☐ 4 bottle caps + 2 skewers + straws (axles)
- ☐ DC motor + propeller
- ☐ AA battery holder with switch

🔧 Build together

TOGETHER

1. Straws taped under the chassis, skewers through, bottle caps as wheels.
2. Mount the motor at the back facing backward with the propeller.
3. Switch on and let it go.

🎮 Play

KID TIME

Distance contest: mark the floor with tape every 50 cm. Who can tweak the car to go furthest?

💬 Talk about it

TOGETHER

Which way does the air go, and which way does the car go?

⭐ Level up

KID TIME

Flip the motor wires. What happens?

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
Chassis	The base of a vehicle that everything else is attached to.	The robot's skeleton.
Axle	The rod that wheels turn on.	The stick the wheels spin on.
DC motor	A motor that spins when connected to a battery. Swap its two wires and it spins the other way.	A spinner that runs on batteries.
Propeller	Spinning blades that push air one way, so the vehicle moves the other way.	Spinning blades that push air.
Switch	Something that opens or closes a gap in a circuit, to turn things off and on.	A little door for electricity: open means off, closed means on.

Week 8 · Fan Car

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

PHASE 1 · CIRCUITS AND MOTORS

9 Two-Motor Tank

~45 min build + short play sessions during the week · Skill: Steering with two motors

Done! ☐

🧰 You need

- ☐ 2 DC motors with wheels (TT gear motors are ideal)
- ☐ 2 battery holders with switches
- ☐ Cardboard chassis
- ☐ A ball caster or a smooth bottle cap at the front

🔧 Build together

TOGETHER

1. Mount one motor on each side at the back.
2. Each motor gets its own battery holder and switch.
3. Both on: forward. Only left on: turns right. Only right on: turns left.

🎮 Play

KID TIME

Kid drives through a cardboard maze using only the two switches.

💬 Talk about it

TOGETHER

How do you make the tank turn if there is no steering wheel?

⭐ Level up

KID TIME

Reverse one motor's wires. Can you spin on the spot?

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
DC motor	A motor that spins when connected to a battery. Swap its two wires and it spins the other way.	A spinner that runs on batteries.
Gear motor	A motor with gears inside: slower but stronger, good for turning wheels.	A slow, strong motor for wheels.
Switch	Something that opens or closes a gap in a circuit, to turn things off and on.	A little door for electricity: open means off, closed means on.
Chassis	The base of a vehicle that everything else is attached to.	The robot's skeleton.
Ball caster	A small ball in a holder that lets the front of a robot slide in any direction.	A rolling ball foot.

Week 9 · Two-Motor Tank

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

10

PHASE 1 · CIRCUITS AND MOTORS

Showcase: The Robot Zoo

~45 min build + short play sessions during the week · Skill: Review

Done! ☐

🧰 You need

- ☐ Everything built so far
- ☐ Phone for filming

🔧 Build together

TOGETHER

1. Set up a 'zoo' on the table with name cards for each robot.
2. Kid picks a favorite and fixes or improves it.

🎮 Play

KID TIME

Kid gives a one-minute tour you film. Send it to family.

💬 Talk about it

TOGETHER

Which robot was hardest to build? Which was most fun?

⭐ Level up

KID TIME

Invent a new animal robot from the parts you have.

Week 10 · Showcase: The Robot Zoo

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____



PHASE 2 · WEEKS 11–20

micro:bit First Steps

A tiny computer that senses and reacts. The programs are ready-made; your kid changes icons, numbers and sounds.

PHASE 2 · MICRO:BIT FIRST STEPS

11 Robot Pet Face

~45 min build + short play sessions during the week · Skill: Inputs and outputs

Done! ☐

🧰 You need

- ☐ BBC micro:bit V2 + USB cable
- ☐ Battery pack (2×AAA)
- ☐ Cardboard pet body

🔧 Build together

TOGETHER

1. Plug in the micro:bit and load this week's program (steps below).
2. Build a cardboard pet with a window for the LED screen.

🎮 Play

KID TIME

Button A = happy, B = sad, A+B = surprised, shake = dizzy. Kid decides what each one means for the pet.

💬 Talk about it

TOGETHER

What are the pet's inputs? What are its outputs?

⭐ Level up

KID TIME

Change the icons. Draw your own face with the LED editor.

💻 Load the program

GROWN-UP JOB

A grown-up loads the program (main.ts) from the website: **https://pandagardenio.github.io/robokids/web/phase-2-microbit-first-steps/week-11-robot-pet-face/**. Each week's page has a big Copy button and the steps.

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
Input	Something that goes into the robot: a button press, a shake, light, sound.	What the robot feels.
Output	Something the robot does: lights, sounds, moving.	What the robot does.
V2	The newer micro:bit, with a speaker, a microphone and a touch-sensitive logo. This project needs it.	The newer micro:bit that can hear and talk.
USB	The cable between the laptop and the micro:bit. It carries messages, not just power.	The cable that lets the laptop talk to the robot.
LED screen	The 25 little lights on the front of the micro:bit that show pictures, numbers and letters.	The robot's face.
Accelerometer	The sensor that feels movement: shaking, tilting, falling.	It feels when you shake or tip the robot.
Icon	A small picture shown on the LED screen, like a heart or a smiley.	A tiny picture made of lights.

Week 11 · Robot Pet Face

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

PHASE 2 · MICRO:BIT FIRST STEPS

12 Shake Dice

~45 min build + short play sessions during the week · Skill: Random numbers and the accelerometer

Done! ☐

🧰 You need

☐ micro:bit + battery pack

🔧 Build together

TOGETHER

1. Load this week's program (steps below). Shake to roll 1–6.

🎮 Play

KID TIME

Use the dice for your favorite board game for the whole week.

💬 Talk about it

TOGETHER

How does the micro:bit know it was shaken?

⭐ Level up

KID TIME

Make a 1–20 dice, or a dice that shows arrows for the next week's robot car game.

💻 Load the program

GROWN-UP JOB

A grown-up loads the program (main.ts) from the website: **https://pandagardenio.github.io/robokids/web/phase-2-microbit-first-steps/week-12-shake-dice/**. Each week's page has a big Copy button and the steps.

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
Random	Picked by chance, so nobody (not even the robot) knows what comes next.	Like rolling a dice.
Accelerometer	The sensor that feels movement: shaking, tilting, falling.	It feels when you shake or tip the robot.

Week 12 · Shake Dice

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

PHASE 2 · MICRO:BIT FIRST STEPS

13 Robot Heartbeat

~45 min build + short play sessions during the week · Skill: Loops and timing

Done! ☐

🧰 You need

☐ micro:bit + battery pack

🔧 Build together

TOGETHER

1. Load this week's program (steps below). The heart beats forever.
2. Button A = faster (excited robot). Button B = slower (sleepy robot).

🎮 Play

KID TIME

Run around and count your own heartbeat, then set the robot to match it.

💬 Talk about it

TOGETHER

What does 'forever' mean for a computer?

⭐ Level up

KID TIME

Add a beep on each beat.

💻 Load the program

GROWN-UP JOB

A grown-up loads the program (main.ts) from the website: **https://pandagardenio.github.io/robokids/web/phase-2-microbit-first-steps/week-13-heartbeat/**. Each week's page has a big Copy button and the steps.

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
Timing	How long the robot waits or does something. In programs it's often in milliseconds: 1000 = 1 second.	How long it waits.
Loop	Instructions that repeat. *Forever* repeats until the power goes off; *repeat 4* does it four times.	Do it again, and again, and again.

Week 13 · Robot Heartbeat

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

14

PHASE 2 · MICRO:BIT FIRST STEPS

Thermometer Robot

~45 min build + short play sessions during the week · Skill: Temperature sensor

Done! ☐

🧰 You need

- ☐ micro:bit + battery pack
- ☐ Glass of cold water and warm water (in a plastic bag for the micro:bit, or just hold it near)

🔧 Build together

TOGETHER

1. Load this week's program (steps below). Press A to see the temperature.

🎮 Play

KID TIME

Temperature hunt: fridge door, sunny window, under a blanket. Kid predicts first: hotter or colder?

💬 Talk about it

TOGETHER

Where was the hottest place in the house?

⭐ Level up

KID TIME

Show a snowflake below 15 °C and a sun above 25 °C.

💻 Load the program

GROWN-UP JOB

A grown-up loads the program (main.ts) from the website: **https://pandagardenio.github.io/robokids/web/phase-2-microbit-first-steps/week-14-thermometer-robot/**. Each week's page has a big Copy button and the steps.

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
Temperature	How hot or cold something is, in degrees Celsius (°C). A comfy room is about 20 °C.	How hot or cold.
Sensor	A part that lets the robot notice the world: light, heat, distance, sound, movement.	The robot's eyes, ears and skin.

Week 14 · Thermometer Robot

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

PHASE 2 · MICRO:BIT FIRST STEPS

15 Compass Robot

~45 min build + short play sessions during the week · Skill: Magnetic sensor

Done! ☐

🧰 You need

- ☐ micro:bit + battery pack
- ☐ Paper with N, E, S, W drawn

🔧 Build together

TOGETHER

1. Load this week's program (steps below). First run asks to 'fill the screen' by tilting: that's calibration.
2. The arrow always points north.

🎮 Play

KID TIME

Hide a toy north of the couch and give compass clues. Then switch roles.

💬 Talk about it

TOGETHER

How does the robot know where north is without eyes?

⭐ Level up

KID TIME

Show the letter N, E, S or W instead of an arrow.

💻 Load the program

GROWN-UP JOB

A grown-up loads the program (main.ts) from the website: **https://pandagardenio.github.io/robokids/web/phase-2-microbit-first-steps/week-15-compass-robot/**. Each week's page has a big Copy button and the steps.

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
Compass	The micro:bit has a sensor that feels the Earth's magnetism, like a compass needle, so it knows where north is.	It always knows which way is north.
Sensor	A part that lets the robot notice the world: light, heat, distance, sound, movement.	The robot's eyes, ears and skin.
Accelerometer	The sensor that feels movement: shaking, tilting, falling.	It feels when you shake or tip the robot.
Calibration	Setting up a sensor so it measures correctly. The compass asks for it the first time.	Teaching the sensor where things are.

Week 15 · Compass Robot

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

PHASE 2 · MICRO:BIT FIRST STEPS

16 Night-Light Robot

~45 min build + short play sessions during the week · Skill: Light sensor

Done! ☐

🧰 You need

- ☐ micro:bit + battery pack
- ☐ Cardboard robot with a window

🔧 Build together

TOGETHER

1. Load this week's program (steps below). The LED screen is also a light sensor.
2. Cover it with a hand: the robot lights up.

🎮 Play

KID TIME

Peekaboo: cover the robot's eyes to wake it up, uncover to put it to sleep.

💬 Talk about it

TOGETHER

How can the same LEDs show light AND measure it?

⭐ Level up

KID TIME

Change how dark it must be before the robot wakes up.

💻 Load the program

GROWN-UP JOB

A grown-up loads the program (main.ts) from the website: **https://pandagardenio.github.io/robokids/web/phase-2-microbit-first-steps/week-16-night-light/**. Each week's page has a big Copy button and the steps.

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
Sensor	A part that lets the robot notice the world: light, heat, distance, sound, movement.	The robot's eyes, ears and skin.
LED screen	The 25 little lights on the front of the micro:bit that show pictures, numbers and letters.	The robot's face.
Light sensor	The micro:bit's LED screen can also measure how bright the room is.	The robot's face can also see light.
LED	A tiny light. It only works one way round: the long leg goes to plus (+).	A tiny light that only works the right way round.

Week 16 · Night-Light Robot

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

17

PHASE 2 · MICRO:BIT FIRST STEPS

Robot Music Box

~45 min build + short play sessions during the week · Skill: Sound and sequences

Done! ☐

🧰 You need

- ☐ micro:bit V2 (has a speaker)

🔧 Build together

TOGETHER

1. Load this week's program (steps below). A plays a built-in melody, B plays the kid's own tune.

🎮 Play

KID TIME

Kid composes a 6-note tune with you (edit the notes in the code). Play it to the family.

💬 Talk about it

TOGETHER

What happens if you change the order of the notes?

⭐ Level up

KID TIME

Make the robot play a different tune when shaken.

💻 Load the program

GROWN-UP JOB

A grown-up loads the program (main.ts) from the website: **https://pandagardenio.github.io/robokids/web/phase-2-microbit-first-steps/week-17-music-box/**. Each week's page has a big Copy button and the steps.

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
Sequence	Things done in a fixed order. Change the order and you get a different result.	First this, then that.
V2	The newer micro:bit, with a speaker, a microphone and a touch-sensitive logo. This project needs it.	The newer micro:bit that can hear and talk.
Code	Another word for the program: the instructions the robot follows.	The robot's instructions.
Accelerometer	The sensor that feels movement: shaking, tilting, falling.	It feels when you shake or tip the robot.

Week 17 · Robot Music Box

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

18

PHASE 2 · MICRO:BIT FIRST STEPS

Rock, Paper, Scissors

~45 min build + short play sessions during the week · Skill: Random choice and if/else

Done! ☐

🧰 You need

☐ micro:bit + battery pack

🔧 Build together

TOGETHER

1. Load this week's program (steps below). Shake to make the robot choose.

🎮 Play

KID TIME

Play against the robot. Keep score on paper for the week.

💬 Talk about it

TOGETHER

Can you predict what the robot will choose?

⭐ Level up

KID TIME

Add a fourth choice (lizard? spock?).

💻 Load the program

GROWN-UP JOB

A grown-up loads the program (main.ts) from the website: **https://pandagardenio.github.io/robokids/web/phase-2-microbit-first-steps/week-18-rock-paper-scissors/**. Each week's page has a big Copy button and the steps.

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
Random	Picked by chance, so nobody (not even the robot) knows what comes next.	Like rolling a dice.
If / else	The program checks something: IF it's true, do one thing, ELSE (otherwise) do another.	If it rains, take an umbrella, else take sunglasses.

Week 18 · Rock, Paper, Scissors

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

PHASE 2 · MICRO:BIT FIRST STEPS

19 Secret Radio Messages

~45 min build + short play sessions during the week · Skill: Radio communication

Done! ☐

🧰 You need

- ☐ 2 micro:bits (borrow one or buy a second)
- ☐ 2 battery packs

🔧 Build together

TOGETHER

1. Load the same program onto both micro:bits (steps below).
2. A sends a heart, B sends a smiley, shake sends a skull for 'danger'.

🎮 Play

KID TIME

Spy game: one person hides in another room. Agree on a code (heart = 'come here', smiley = 'all good').

💬 Talk about it

TOGETHER

How did the message get there with no wire?

⭐ Level up

KID TIME

Change the radio group number so nobody else can listen.

💻 Load the program

GROWN-UP JOB

A grown-up loads the program (main.ts) from the website: **https://pandagardenio.github.io/robokids/web/phase-2-microbit-first-steps/week-19-secret-radio/**. Each week's page has a big Copy button and the steps.

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
Radio	micro:bits can send short messages to each other through the air, no wires needed, across a house.	Invisible messages through the air.
Accelerometer	The sensor that feels movement: shaking, tilting, falling.	It feels when you shake or tip the robot.
Radio group	A channel number. Only micro:bits set to the same number hear each other.	A secret channel number.

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

PHASE 2 · MICRO:BIT FIRST STEPS

20 Showcase: Robot Badge

~45 min build + short play sessions during the week · Skill: Review

Done! ☐

🧰 You need

- ☐ micro:bit + battery pack
- ☐ Lanyard or safety pin + cardboard holder

🔧 Build together

TOGETHER

1. Combine favorite features: name scroll, pet face, dice.

🎮 Play

KID TIME

Kid wears the badge for a day and demos it to everyone they meet.

💬 Talk about it

TOGETHER

Which feature do people like most?

⭐ Level up

KID TIME

Add their own idea to the badge.

💻 Load the program

GROWN-UP JOB

A grown-up loads the program (main.ts) from the website: **https://pandagardenio.github.io/robokids/web/phase-2-microbit-first-steps/week-20-showcase-badge/**. Each week's page has a big Copy button and the steps.

Week 20 · Showcase: Robot Badge

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____



PHASE 3 · WEEKS 21–30

The Robot Car

Motors controlled by code. Build the Cutebot together, then make it drive, light up, follow lines and dodge walls.

PHASE 3 · THE ROBOT CAR

21 Build the Robot Car

~45 min build + short play sessions during the week · Skill: Parts of a robot

Done! ☐

🧰 You need

- ☐ Elecfreaks Cutebot kit (or similar micro:bit car)
- ☐ micro:bit
- ☐ 3×AAA batteries (in the kit's holder)
- ☐ Small screwdriver

🔧 Build together

TOGETHER

1. Unbox together and name every part: wheels, motors, sensors, lights, brain (micro:bit).
2. Insert batteries (parent), slot the micro:bit in, screen facing forward.
3. Load `main.ts`: the car says hello, lights up, and wiggles.

🎮 Play

KID TIME

Name the robot and make a cardboard name plate.

💬 Talk about it

TOGETHER

Which part is the robot's brain? Its eyes? Its legs?

⭐ Level up

KID TIME

Draw the robot in the Robot Journal and label the parts.

💻 Load the program

GROWN-UP JOB

A grown-up loads the program (main.ts) from the website: **https://pandagardenio.github.io/robokids/web/phase-3-robot-car/week-21-build-the-car/**. Each week's page has a big Copy button and the steps.

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
Cutebot	The robot car kit this project uses. The micro:bit slots in as its brain.	Our robot car.
Sensor	A part that lets the robot notice the world: light, heat, distance, sound, movement.	The robot's eyes, ears and skin.

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

PHASE 3 · THE ROBOT CAR

22 Forward, Stop, Back

~45 min build + short play sessions during the week · Skill: Motor speed and time

Done! ☐

🧰 You need

- ☐ Robot car
- ☐ Masking tape for a start and finish line

🔧 Build together

TOGETHER

1. Load this week's program (steps below). A = forward 1 s, B = backward 1 s.

🎮 Play

KID TIME

Parking game: tape a parking spot. Kid changes the time in the code to park exactly.

💬 Talk about it

TOGETHER

What changes when you make the number bigger?

⭐ Level up

KID TIME

Change the speed instead of the time. Is it the same?

💻 Load the program

GROWN-UP JOB

A grown-up loads the program (main.ts) from the website: **https://pandagardenio.github.io/robokids/web/phase-3-robot-car/week-22-forward-stop-back/**. Each week's page has a big Copy button and the steps.

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
Motor speed	How fast the wheels turn, from 0 (stopped) to 100 (full speed). Negative numbers drive backwards.	How fast the wheels spin.
Timing	How long the robot waits or does something. In programs it's often in milliseconds: 1000 = 1 second.	How long it waits.
Code	Another word for the program: the instructions the robot follows.	The robot's instructions.

Week 22 · Forward, Stop, Back

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

PHASE 3 · THE ROBOT CAR

23 Square Dance

~45 min build + short play sessions during the week · Skill: Loops

Done! ☐

🧰 You need

- ☐ Robot car
- ☐ Smooth floor

🔧 Build together

TOGETHER

1. Load this week's program (steps below). Press A: the car drives a square using a 'repeat 4' loop.

🎮 Play

KID TIME

Tune the turn time until the square closes perfectly. Then make a triangle (repeat 3).

💬 Talk about it

TOGETHER

Why use 'repeat 4' instead of writing it four times?

⭐ Level up

KID TIME

Make a star or a dance with music.

💻 Load the program

GROWN-UP JOB

A grown-up loads the program (main.ts) from the website: **https://pandagardenio.github.io/robokids/web/phase-3-robot-car/week-23-square-dance/**. Each week's page has a big Copy button and the steps.

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
Loop	Instructions that repeat. *Forever* repeats until the power goes off; *repeat 4* does it four times.	Do it again, and again, and again.
Timing	How long the robot waits or does something. In programs it's often in milliseconds: 1000 = 1 second.	How long it waits.

Week 23 · Square Dance

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

PHASE 3 · THE ROBOT CAR

24 Button Steering

~45 min build + short play sessions during the week · Skill: Events

Done! ☐

🧰 You need

- ☐ Robot car

🔧 Build together

TOGETHER

1. Load this week's program (steps below). A = turn left, B = turn right, A+B = forward, shake = stop.

🎮 Play

KID TIME

Slalom around shoes on the floor.

💬 Talk about it

TOGETHER

How does the robot know which button was pressed?

⭐ Level up

KID TIME

Add a 'spin' move when you press the logo (V2 touch logo).

💻 Load the program

GROWN-UP JOB

A grown-up loads the program (main.ts) from the website: **https://pandagardenio.github.io/robokids/web/phase-3-robot-car/week-24-button-steering/**. Each week's page has a big Copy button and the steps.

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
Event	Something that happens, like a button press or a shake, and the instructions that run when it does.	When this happens, do that.
Accelerometer	The sensor that feels movement: shaking, tilting, falling.	It feels when you shake or tip the robot.
V2	The newer micro:bit, with a speaker, a microphone and a touch-sensitive logo. This project needs it.	The newer micro:bit that can hear and talk.

Week 24 · Button Steering

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

PHASE 3 · THE ROBOT CAR

25 Disco Headlights

~45 min build + short play sessions during the week · Skill: Colors and RGB

Done! ☐

🧰 You need

- ☐ Robot car
- ☐ Dark room

🔧 Build together

TOGETHER

1. Load this week's program (steps below). Each button press changes the headlight color.

🎮 Play

KID TIME

Disco: robot flashes colors to music while dancing.

💬 Talk about it

TOGETHER

How can three colors (red, green, blue) make all the others?

⭐ Level up

KID TIME

Choose colors for 'happy', 'angry' and 'sleepy'.

💻 Load the program

GROWN-UP JOB

A grown-up loads the program (main.ts) from the website: **https://pandagardenio.github.io/robokids/web/phase-3-robot-car/week-25-headlights/**. Each week's page has a big Copy button and the steps.

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
RGB	Red, Green, Blue. Mixing these three colors of light makes every other color.	Three colors that mix into all the others.

Week 25 · Disco Headlights

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

PHASE 3 · THE ROBOT CAR

26 Line Follower

~45 min build + short play sessions during the week · Skill: Sensors + if/else

Done! ☐

🧰 You need

- ☐ Robot car
- ☐ Black electrical tape (2 cm wide)
- ☐ Big white paper or light floor

🔧 Build together

TOGETHER

1. Make a closed loop track with gentle curves.
2. Load this week's program (steps below). Put the robot on the line and press A.

🎮 Play

KID TIME

Kid designs new tracks each day. Add a 'station' where the robot should stop (you can add that in code).

💬 Talk about it

TOGETHER

How does the robot know it's leaving the line?

⭐ Level up

KID TIME

Make a figure-8 track. Does it still work?

💻 Load the program

GROWN-UP JOB

A grown-up loads the program (main.ts) from the website: **https://pandagardenio.github.io/robokids/web/phase-3-robot-car/week-26-line-follower/**. Each week's page has a big Copy button and the steps.

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
If / else	The program checks something: IF it's true, do one thing, ELSE (otherwise) do another.	If it rains, take an umbrella, else take sunglasses.
Line sensor	Two small sensors under the car that see whether the floor below is dark or light.	The robot's eyes looking at the floor.
Code	Another word for the program: the instructions the robot follows.	The robot's instructions.

Week 26 · Line Follower

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

PHASE 3 · THE ROBOT CAR

27 Obstacle Avoider

~45 min build + short play sessions during the week · Skill: Distance sensor

Done! ☐

🧰 You need

- ☐ Robot car with ultrasonic sensor (included with Cutebot)
- ☐ Boxes and books as walls

🔧 Build together

TOGETHER

1. Load this week's program (steps below). The robot drives until something is closer than 15 cm, then backs up and turns.

🎮 Play

KID TIME

Escape the cave: build a box maze and see how long it takes to get out.

💬 Talk about it

TOGETHER

How does the robot 'see' without eyes? (Like a bat.)

⭐ Level up

KID TIME

Change 15 cm to 30 cm. Is it more careful now?

💻 Load the program

GROWN-UP JOB

A grown-up loads the program (main.ts) from the website: **https://pandagardenio.github.io/robokids/web/phase-3-robot-car/week-27-obstacle-avoider/**. Each week's page has a big Copy button and the steps.

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
Ultrasonic sensor	The two round 'eyes' on the front of the Cutebot. They send out a sound too high to hear and time the echo to measure distance.	The robot squeaks and listens for the echo, like a bat.
Sensor	A part that lets the robot notice the world: light, heat, distance, sound, movement.	The robot's eyes, ears and skin.
Cutebot	The robot car kit this project uses. The micro:bit slots in as its brain.	Our robot car.

Week 27 · Obstacle Avoider

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

PHASE 3 · THE ROBOT CAR

28 Tilt Remote Control

~45 min build + short play sessions during the week · Skill: Radio + accelerometer

Done! ☐

🧰 You need

- ☐ Robot car
- ☐ Second micro:bit as remote

🔧 Build together

TOGETHER

1. Load `remote.ts` onto the hand micro:bit and `main.ts` onto the car.
2. Tilt forward to drive, tilt sideways to turn, press A to stop.

🎮 Play

KID TIME

Treasure hunt: drive the car to pick up paper 'treasure' with a cardboard scoop.

💬 Talk about it

TOGETHER

How does tilting your hand move the car?

⭐ Level up

KID TIME

Press B to honk.

💻 Load the program

GROWN-UP JOB

A grown-up loads the program (main.ts, remote.ts) from the website: **https://pandagardenio.github.io/robokids/web/phase-3-robot-car/week-28-tilt-remote/**. Each week's page has a big Copy button and the steps.

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
Radio	micro:bits can send short messages to each other through the air, no wires needed, across a house.	Invisible messages through the air.
Accelerometer	The sensor that feels movement: shaking, tilting, falling.	It feels when you shake or tip the robot.

Week 28 · Tilt Remote Control

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

PHASE 3 · THE ROBOT CAR

29 Delivery Mission

~45 min build + short play sessions during the week · Skill: Planning a program

Done! ☐

🧰 You need

- ☐ Robot car
- ☐ Cardboard houses
- ☐ Small 'parcels' (paper squares)

🔧 Build together

TOGETHER

1. Set up 3 houses. Kid plans the route on paper with arrows.
2. Translate the arrows into the moves list in the code together.

🎮 Play

KID TIME

Robot delivers a parcel to each house. Did it arrive?

💬 Talk about it

TOGETHER

If the robot misses a house, where is the bug: in the plan or in the code?

⭐ Level up

KID TIME

Add a pickup sound at each house.

💻 Load the program

GROWN-UP JOB

A grown-up loads the program (main.ts) from the website: **https://pandagardenio.github.io/robokids/web/phase-3-robot-car/week-29-delivery-mission/**. Each week's page has a big Copy button and the steps.

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
List	A set of items in order. The program goes through them one by one.	Like a shopping list for the robot.
Code	Another word for the program: the instructions the robot follows.	The robot's instructions.
Bug	A mistake that makes a robot do the wrong thing. Finding and fixing it is called debugging.	A mistake we get to find and fix.

Week 29 · Delivery Mission

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

PHASE 3 · THE ROBOT CAR

30 Showcase: Robot Olympics

~45 min build + short play sessions during the week · Skill: Review

Done! ☐

🧰 You need

- ☐ Robot car
- ☐ Track, maze, parking spot from earlier weeks
- ☐ Bottle-cap medals

🔧 Build together

TOGETHER

1. Load the menu program (steps below): A cycles through events, B starts.

🎮 Play

KID TIME

Three events: line race (timed), maze escape, parking. Family is the jury.

💬 Talk about it

TOGETHER

Which event was hardest for the robot?

⭐ Level up

KID TIME

Invent a fourth event.

💻 Load the program

GROWN-UP JOB

A grown-up loads the program (main.ts) from the website: **https://pandagardenio.github.io/robokids/web/phase-3-robot-car/week-30-showcase-robot-olympics/**. Each week's page has a big Copy button and the steps.

Week 30 · Showcase: Robot Olympics

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____



PHASE 4 · WEEKS 31–40

Smarter Robots

Combining sensors into robot behaviors: arms, moods, guards, games between two robots.

PHASE 4 · SMARTER ROBOTS

31 Waving Robot Arm

~45 min build + short play sessions during the week · Skill: Servo motors

Done! ☐

🧰 You need

- ☐ Robot car
- ☐ Small servo (SG90) plugged into Cutebot S1
- ☐ Cardboard arm + hand

🔧 Build together

TOGETHER

1. Attach the cardboard arm to the servo horn.
2. Load this week's program (steps below). A = wave hello, B = high five position.

🎮 Play

KID TIME

Robot greets everyone who comes home.

💬 Talk about it

TOGETHER

How is a servo different from a wheel motor?

⭐ Level up

KID TIME

Make it wave faster when it's 'excited'.

💻 Load the program

GROWN-UP JOB

A grown-up loads the program (main.ts) from the website: **https://pandagardenio.github.io/robokids/web/phase-4-smarter-robots/week-31-waving-arm/**. Each week's page has a big Copy button and the steps.

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
Servo	A motor that turns to an exact angle and holds still there, instead of spinning round and round.	A motor that points instead of spins.
Cutebot	The robot car kit this project uses. The micro:bit slots in as its brain.	Our robot car.
S1	A socket on the Cutebot labeled S1 where you plug in a servo.	The servo's plug spot.
Servo horn	The small plastic arm that clips onto the servo and moves with it.	The servo's little arm.

Week 31 · Waving Robot Arm

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

PHASE 4 · SMARTER ROBOTS

32 Bulldozer

~45 min build + short play sessions during the week · Skill: Pushing and power

Done! ☐

🧰 You need

- ☐ Robot car
- ☐ Cardboard plow taped to the front
- ☐ Duplo or foam blocks

🔧 Build together

TOGETHER

1. Load this week's program (steps below). The bulldozer pushes forward until it sees a wall, then backs up.

🎮 Play

KID TIME

Clean-up time: push blocks into a 'garage' (shoebox on its side).

💬 Talk about it

TOGETHER

Is the robot faster pushing or not pushing? Why?

⭐ Level up

KID TIME

Use the servo arm as a scoop.

💻 Load the program

GROWN-UP JOB

A grown-up loads the program (main.ts) from the website: **https://pandagardenio.github.io/robokids/web/phase-4-smarter-robots/week-32-bulldozer/**. Each week's page has a big Copy button and the steps.

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
Ultrasonic sensor	The two round 'eyes' on the front of the Cutebot. They send out a sound too high to hear and time the echo to measure distance.	The robot squeaks and listens for the echo, like a bat.
Servo	A motor that turns to an exact angle and holds still there, instead of spinning round and round.	A motor that points instead of spins.

Week 32 · Bulldozer

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

PHASE 4 · SMARTER ROBOTS

33 Clap-Activated Robot

~45 min build + short play sessions during the week · Skill: Sound sensor

Done! ☐

🧰 You need

- ☐ Robot car with micro:bit V2 (microphone)

🔧 Build together

TOGETHER

1. Load this week's program (steps below). One loud clap = go / stop. The LED shows how loud the room is.

🎮 Play

KID TIME

Freeze dance with the robot: clap to start and stop.

💬 Talk about it

TOGETHER

Does the robot hear words or only loudness?

⭐ Level up

KID TIME

Two claps = spin (count claps in a short time window).

💻 Load the program

GROWN-UP JOB

A grown-up loads the program (main.ts) from the website: **https://pandagardenio.github.io/robokids/web/phase-4-smarter-robots/week-33-clap-robot/**. Each week's page has a big Copy button and the steps.

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
Microphone	The micro:bit V2 can measure how loud it is. It hears loudness, not words.	The robot's ear. It hears loud and quiet, not words.
Sensor	A part that lets the robot notice the world: light, heat, distance, sound, movement.	The robot's eyes, ears and skin.
V2	The newer micro:bit, with a speaker, a microphone and a touch-sensitive logo. This project needs it.	The newer micro:bit that can hear and talk.
LED	A tiny light. It only works one way round: the long leg goes to plus (+).	A tiny light that only works the right way round.

Week 33 · Clap-Activated Robot

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

PHASE 4 · SMARTER ROBOTS

34 Follow-Me Robot

~45 min build + short play sessions during the week · Skill: Keeping distance

Done! ☐

🧰 You need

- ☐ Robot car
- ☐ A flat book as the 'leader'

🔧 Build together

TOGETHER

1. Load this week's program (steps below). The robot tries to stay about 20 cm from the book.

🎮 Play

KID TIME

Walk the robot like a puppy around the room.

💬 Talk about it

TOGETHER

What does the robot do if you come too close?

⭐ Level up

KID TIME

Change 20 cm to 40 cm: a shy puppy.

💻 Load the program

GROWN-UP JOB

A grown-up loads the program (main.ts) from the website: **https://pandagardenio.github.io/robokids/web/phase-4-smarter-robots/week-34-follow-me/**. Each week's page has a big Copy button and the steps.

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
Ultrasonic sensor	The two round 'eyes' on the front of the Cutebot. They send out a sound too high to hear and time the echo to measure distance.	The robot squeaks and listens for the echo, like a bat.

Week 34 · Follow-Me Robot

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

PHASE 4 · SMARTER ROBOTS

35 Guard Robot

~45 min build + short play sessions during the week · Skill: Alarms and states

Done! ☐

🧰 You need

- ☐ Robot car
- ☐ A treasure (toy) to guard

🔧 Build together

TOGETHER

1. Load this week's program (steps below). Press A to arm. If something comes within 30 cm, alarm lights and sound.

🎮 Play

KID TIME

Sneak-up game: try to steal the treasure without triggering the guard.

💬 Talk about it

TOGETHER

What does 'armed' mean for the robot?

⭐ Level up

KID TIME

Make the guard turn toward the thief before sounding.

💻 Load the program

GROWN-UP JOB

A grown-up loads the program (main.ts) from the website: **https://pandagardenio.github.io/robokids/web/phase-4-smarter-robots/week-35-guard-robot/**. Each week's page has a big Copy button and the steps.

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
State	The mode the robot is in right now, like sleeping, guarding or alarm. It acts differently in each.	The robot's mood right now.
Armed	Switched on and watching, ready to sound the alarm.	On guard.
Ultrasonic sensor	The two round 'eyes' on the front of the Cutebot. They send out a sound too high to hear and time the echo to measure distance.	The robot squeaks and listens for the echo, like a bat.

Week 35 · Guard Robot

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

PHASE 4 · SMARTER ROBOTS

36 Robot Moods

~45 min build + short play sessions during the week · Skill: Combining sensors

Done! ☐

🧰 You need

- ☐ Robot car

🔧 Build together

TOGETHER

1. Load this week's program (steps below). Dark = sleepy (dim, slow). Shake = dizzy (spins). Loud = scared (backs away). Button A = happy dance.

🎮 Play

KID TIME

Kid tells a story and makes the robot feel each mood.

💬 Talk about it

TOGETHER

How many sensors is the robot using at once?

⭐ Level up

KID TIME

Add a new mood with a new sensor (temperature: cold = shivering).

💻 Load the program

GROWN-UP JOB

A grown-up loads the program (main.ts) from the website: **https://pandagardenio.github.io/robokids/web/phase-4-smarter-robots/week-36-robot-moods/**. Each week's page has a big Copy button and the steps.

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
Sensor	A part that lets the robot notice the world: light, heat, distance, sound, movement.	The robot's eyes, ears and skin.
Temperature	How hot or cold something is, in degrees Celsius (°C). A comfy room is about 20 °C.	How hot or cold.

Week 36 · Robot Moods

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

PHASE 4 · SMARTER ROBOTS

37 Hot and Cold Treasure Hunt

~45 min build + short play sessions during the week · Skill: Signal strength

Done! ☐

🧰 You need

- ☐ 2 micro:bits (one hidden as the 'treasure beacon')
- ☐ Battery packs

🔧 Build together

TOGETHER

1. Load `beacon.ts` onto the hidden micro:bit and `main.ts` onto the seeker (car or handheld).

🎮 Play

KID TIME

Hide the beacon. The seeker's screen fills up the closer it gets. Find it!

💬 Talk about it

TOGETHER

Why does the signal get stronger as you get closer?

⭐ Level up

KID TIME

Hide it inside a metal pot. What happens?

💻 Load the program

GROWN-UP JOB

A grown-up loads the program (main.ts, beacon.ts) from the website: **https://pandagardenio.github.io/robokids/web/phase-4-smarter-robots/week-37-hot-cold-treasure/**. Each week's page has a big Copy button and the steps.

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
Signal strength	How strong a radio message is when it arrives. Closer usually means stronger.	Louder when close, quieter when far.
Beacon	A micro:bit that keeps sending the same radio message so another one can find it.	A hidden micro:bit that keeps calling 'I'm here!'.
LED screen	The 25 little lights on the front of the micro:bit that show pictures, numbers and letters.	The robot's face.

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

PHASE 4 · SMARTER ROBOTS

38 Drawing Robot

~45 min build + short play sessions during the week · Skill: Geometry with code

Done! ☐

🧰 You need

- ☐ Robot car
- ☐ Felt-tip marker taped to the back center
- ☐ Large paper

🔧 Build together

TOGETHER

1. Load this week's program (steps below). A = square, B = triangle, A+B = flower (loop of loops).

🎮 Play

KID TIME

Make an art gallery of robot drawings.

💬 Talk about it

TOGETHER

How many turns make a full circle?

⭐ Level up

KID TIME

Write their initial with the robot.

💻 Load the program

GROWN-UP JOB

A grown-up loads the program (main.ts) from the website: **https://pandagardenio.github.io/robokids/web/phase-4-smarter-robots/week-38-drawing-robot/**. Each week's page has a big Copy button and the steps.

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
Geometry	The maths of shapes. A full turn is 360°; a square has four turns of 90°.	Shapes and turns.
Loop	Instructions that repeat. *Forever* repeats until the power goes off; *repeat 4* does it four times.	Do it again, and again, and again.

Week 38 · Drawing Robot

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

PHASE 4 · SMARTER ROBOTS

39 Robot Tag

~45 min build + short play sessions during the week · Skill: Two robots talking

Done! ☐

🧰 You need

- ☐ 2 micro:bits (one on the car, one handheld 'tagger')

🔧 Build together

TOGETHER

1. Load `main.ts` onto the car and `tagger.ts` onto the handheld.
2. The car runs around; when the tagger is very close (strong radio signal) the car is 'it' and flashes red.

🎮 Play

KID TIME

Chase the robot around the living room with the tagger.

💬 Talk about it

TOGETHER

How does the robot know you're close without seeing you?

⭐ Level up

KID TIME

Swap: the robot chases you.

💻 Load the program

GROWN-UP JOB

A grown-up loads the program (main.ts, tagger.ts) from the website: **https://pandagardenio.github.io/robokids/web/phase-4-smarter-robots/week-39-robot-tag/**. Each week's page has a big Copy button and the steps.

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
Radio	micro:bits can send short messages to each other through the air, no wires needed, across a house.	Invisible messages through the air.
Signal strength	How strong a radio message is when it arrives. Closer usually means stronger.	Louder when close, quieter when far.

Week 39 · Robot Tag

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

PHASE 4 · SMARTER ROBOTS

40 Showcase: Smart Robot Show

~45 min build + short play sessions during the week · Skill: Review

Done! ☐

🧰 You need

- ☐ Robot car
- ☐ Props from weeks 31–39

🔧 Build together

TOGETHER

1. Kid picks three favorite behaviors. Load the combined show program (steps below).

🎮 Play

KID TIME

Ticketed show for the family (paper tickets, popcorn).

💬 Talk about it

TOGETHER

If you could add one more sensor, what would it be?

⭐ Level up

KID TIME

Kid presents and explains one behavior in their own words.

💻 Load the program

GROWN-UP JOB

A grown-up loads the program (main.ts) from the website: **https://pandagardenio.github.io/robokids/web/phase-4-smarter-robots/week-40-showcase-smart-show/**. Each week's page has a big Copy button and the steps.

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
Behavior	One thing the robot does in response to something, like 'back away when it's loud'.	Something the robot does when something happens.
Sensor	A part that lets the robot notice the world: light, heat, distance, sound, movement.	The robot's eyes, ears and skin.

Week 40 · Showcase: Smart Robot Show

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____



PHASE 5 · WEEKS 41–50

Robots That Learn

Your kid trains AI models by showing examples; the robot acts on what the model sees and hears.

41

PHASE 5 · ROBOTS THAT LEARN

Train Your First AI

~45 min build + short play sessions during the week · Skill: Learning from examples

Done! ☐

🧰 You need

- ☐ Laptop with webcam and Chrome/Edge
- ☐ teachablemachine.withgoogle.com (free)

🔧 Build together

TOGETHER

1. Open Teachable Machine → Image Project. Make two classes: `thumbsup` and `thumbsdown`.
2. Kid records ~50 examples of each. Train.
3. Test: the bars show what the AI thinks.

🎮 Play

KID TIME

Trick the AI: half thumbs, gloves, a toy's hand. When does it fail?

💬 Talk about it

TOGETHER

How did the AI learn without anyone writing rules?

⭐ Level up

KID TIME

Add a third class: `nothing` (empty background). Does it get better?

💻 Load the program

GROWN-UP JOB

Nothing to load this week, only training. Keep the model: you'll use it next week.

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
Training	Showing the AI lots of examples so it learns to tell things apart.	Teaching the AI by showing it.
AI	Artificial intelligence: a computer program that learns from examples instead of following rules someone wrote.	A computer that learns by looking at examples.
Examples / data	The photos or sounds you show the AI. More, varied examples make a smarter AI; few or similar ones make a confused one.	The pictures we show the AI to learn from.
Teachable Machine	A free Google website where you teach an AI by showing it examples with the webcam or microphone.	The website where we teach the computer.
Class	One of the things the AI learns to tell apart, like `thumbsup` or `teddy`. Names must match the week's page exactly.	One thing the AI learns to spot.

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

PHASE 5 · ROBOTS THAT LEARN

42 The AI Bridge

~45 min build + short play sessions during the week · Skill: Connecting AI to a robot

Done! ☐

🧰 You need

- ☐ Laptop + Chrome/Edge
- ☐ micro:bit + USB cable
- ☐ Last week's model (Export → Upload → copy the link)

🔧 Build together

TOGETHER

1. Load `main.ts` onto the micro:bit and keep it plugged in by USB.
2. Close MakeCode, open the AI bridge, paste the model link, click **Connect micro:bit**, then **Start AI** (full steps).
3. The micro:bit shows ✓ or × depending on your thumb.

🎮 Play

KID TIME

Silent game: answer yes/no questions with thumbs only; the robot shows the answer.

💬 Talk about it

TOGETHER

Who decides: the AI or the micro:bit? (Both: AI sees, micro:bit acts.)

⭐ Level up

KID TIME

Make a happy and sad face instead of ✓ and ×.

💻 Load the program

GROWN-UP JOB

A grown-up loads the program (main.ts) from the website: **https://pandagardenio.github.io/robokids/web/phase-5-robots-that-learn/week-42-ai-bridge/**. Each week's page has a big Copy button and the steps.

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
AI	Artificial intelligence: a computer program that learns from examples instead of following rules someone wrote.	A computer that learns by looking at examples.
AI bridge	A page from this project that passes what the AI sees or hears to the micro:bit. See the AI bridge guide.	The messenger between the AI and the robot.
USB	The cable between the laptop and the micro:bit. It carries messages, not just power.	The cable that lets the laptop talk to the robot.
Model	What the AI ends up with after training: its 'memory' of what it learned. Exporting it gives you a link.	The AI's memory.
Export	In Teachable Machine: saving the trained model online so you get a link to paste into the AI bridge.	Sending the AI's memory to the internet so the robot can use it.

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

PHASE 5 · ROBOTS THAT LEARN

43 Gesture-Controlled Car

~45 min build + short play sessions during the week · Skill: AI → action

Done! ☐

🧰 You need

- ☐ Laptop + webcam
- ☐ Robot car
- ☐ Second micro:bit plugged into the laptop as radio relay

🔧 Build together

TOGETHER

1. Train classes `go`, `stop`, `left`, `right`, `nothing` (hand gestures the kid invents).
2. Load `relay.ts` onto the USB micro:bit, `main.ts` to the car.
3. Bridge → relay → radio → car.

🎮 Play

KID TIME

Drive the car through the maze using only hand gestures.

💬 Talk about it

TOGETHER

What happens when the AI is unsure?

⭐ Level up

KID TIME

Add a `dance` gesture.

💻 Load the program

GROWN-UP JOB

A grown-up loads the program (main.ts, relay.ts) from the website: **https://pandagardenio.github.io/robokids/web/phase-5-robots-that-learn/week-43-gesture-car/**. Each week's page has a big Copy button and the steps.

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
AI	Artificial intelligence: a computer program that learns from examples instead of following rules someone wrote.	A computer that learns by looking at examples.
Radio	micro:bits can send short messages to each other through the air, no wires needed, across a house.	Invisible messages through the air.
Relay	A micro:bit in the middle that passes messages on: it gets them from the laptop by USB and sends them to the car by radio.	A messenger in the middle.
Training	Showing the AI lots of examples so it learns to tell things apart.	Teaching the AI by showing it.
Class	One of the things the AI learns to tell apart, like `thumbsup` or `teddy`. Names must match the week's page exactly.	One thing the AI learns to spot.
USB	The cable between the laptop and the micro:bit. It carries messages, not just power.	The cable that lets the laptop talk to the robot.
AI bridge	A page from this project that passes what the AI sees or hears to the micro:bit. See the AI bridge guide.	The messenger between the AI and the robot.
Confidence	How sure the AI is about its guess. The bridge only tells the robot when the AI is sure enough.	How sure the AI is.

Week 43 · Gesture-Controlled Car

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

PHASE 5 · ROBOTS THAT LEARN

44 Clap and Whistle AI

~45 min build + short play sessions during the week · Skill: Training with sound

Done! ☐

🧰 You need

- ☐ Laptop with mic
- ☐ Robot car + USB relay micro:bit (from week 43)

🔧 Build together

TOGETHER

1. Teachable Machine → Audio Project. Classes: `Background Noise` (required), `clap`, `whistle`.
2. Export the model, paste the link into the AI bridge and choose **Audio**.

🎮 Play

KID TIME

Clap = dance, whistle = come back. Play 'robot dog'.

💬 Talk about it

TOGETHER

How is this different from week 33 (just loudness)?

⭐ Level up

KID TIME

Add a `snap` sound for spinning.

💻 Load the program

GROWN-UP JOB

A grown-up loads the program (main.ts) from the website: **https://pandagardenio.github.io/robokids/web/phase-5-robots-that-learn/week-44-clap-and-whistle-ai/**. Each week's page has a big Copy button and the steps.

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
AI	Artificial intelligence: a computer program that learns from examples instead of following rules someone wrote.	A computer that learns by looking at examples.
Training	Showing the AI lots of examples so it learns to tell things apart.	Teaching the AI by showing it.
Microphone	The micro:bit V2 can measure how loud it is. It hears loudness, not words.	The robot's ear. It hears loud and quiet, not words.
USB	The cable between the laptop and the micro:bit. It carries messages, not just power.	The cable that lets the laptop talk to the robot.
Relay	A micro:bit in the middle that passes messages on: it gets them from the laptop by USB and sends them to the car by radio.	A messenger in the middle.
Teachable Machine	A free Google website where you teach an AI by showing it examples with the webcam or microphone.	The website where we teach the computer.
Background Noise	A class of normal room sounds in a Teachable Machine audio project, so the AI knows what 'nothing special' sounds like.	What quiet sounds like.
Export	In Teachable Machine: saving the trained model online so you get a link to paste into the AI bridge.	Sending the AI's memory to the internet so the robot can use it.
Model	What the AI ends up with after training: its 'memory' of what it learned. Exporting it gives you a link.	The AI's memory.
AI bridge	A page from this project that passes what the AI sees or hears to the micro:bit. See the AI bridge guide.	The messenger between the AI and the robot.

Week 44 · Clap and Whistle AI

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

PHASE 5 · ROBOTS THAT LEARN

45 Toy Sorter

~45 min build + short play sessions during the week · Skill: Categories

Done! ☐

🧰 You need

- ☐ Laptop + webcam
- ☐ 3 kinds of toys (e.g. teddy, car, ball)
- ☐ micro:bit by USB

🔧 Build together

TOGETHER

1. Train classes `teddy`, `car`, `ball`, `nothing`.
2. Load `main.ts`. The micro:bit shows an icon and plays a sound for each toy.

🎮 Play

KID TIME

Toy parade: kid shows toys one by one; the robot names them. Then mystery toys it never saw.

💬 Talk about it

TOGETHER

What does the AI do with a toy it never learned?

⭐ Level up

KID TIME

Add a 4th toy and retrain.

💻 Load the program

GROWN-UP JOB

A grown-up loads the program (main.ts) from the website: **https://pandagardenio.github.io/robokids/web/phase-5-robots-that-learn/week-45-toy-sorter/**. Each week's page has a big Copy button and the steps.

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
USB	The cable between the laptop and the micro:bit. It carries messages, not just power.	The cable that lets the laptop talk to the robot.
Training	Showing the AI lots of examples so it learns to tell things apart.	Teaching the AI by showing it.
Class	One of the things the AI learns to tell apart, like `thumbsup` or `teddy`. Names must match the week's page exactly.	One thing the AI learns to spot.
Icon	A small picture shown on the LED screen, like a heart or a smiley.	A tiny picture made of lights.
AI	Artificial intelligence: a computer program that learns from examples instead of following rules someone wrote.	A computer that learns by looking at examples.

Week 45 · Toy Sorter

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

PHASE 5 · ROBOTS THAT LEARN

46 The Bad Data Experiment

~45 min build + short play sessions during the week · Skill: Data quality

Done! ☐

🧰 You need

- ☐ Week 45 setup

🔧 Build together

TOGETHER

1. Train a 'bad' model: only 3 photos per class, in the dark, all from one angle.
2. Train a 'good' model: 50+ photos, lights on, many angles.
3. Test both with the same toys.

🎮 Play

KID TIME

Robot quiz: which model wins more rounds? Kid keeps score.

💬 Talk about it

TOGETHER

Why did the bad robot get confused?

⭐ Level up

KID TIME

Train with only teddies wearing hats. Then show a teddy without a hat.

💻 Load the program

GROWN-UP JOB

A grown-up loads the program (main.ts) from the website: **https://pandagardenio.github.io/robokids/web/phase-5-robots-that-learn/week-46-bad-data-experiment/**. Each week's page has a big Copy button and the steps.

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
Training	Showing the AI lots of examples so it learns to tell things apart.	Teaching the AI by showing it.
Model	What the AI ends up with after training: its 'memory' of what it learned. Exporting it gives you a link.	The AI's memory.
Examples / data	The photos or sounds you show the AI. More, varied examples make a smarter AI; few or similar ones make a confused one.	The pictures we show the AI to learn from.
Class	One of the things the AI learns to tell apart, like `thumbsup` or `teddy`. Names must match the week's page exactly.	One thing the AI learns to spot.

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

PHASE 5 · ROBOTS THAT LEARN

47 AI Story Robot

~45 min build + short play sessions during the week · Skill: AI helps create

Done! ☐

🧰 You need

- ☐ Robot car
- ☐ Laptop (parent uses an AI assistant)

🔧 Build together

TOGETHER

1. Kid invents a hero robot and a problem. Parent asks an AI for a 5-scene bedtime story with one robot action per scene.
2. Edit the `SCENES` list in the code to match: each scene = a move + face + sound.
3. Each A press plays the next scene while you read.

🎮 Play

KID TIME

Bedtime performance: you read, the robot acts.

💬 Talk about it

TOGETHER

Which ideas came from you, and which from the AI?

⭐ Level up

KID TIME

Kid writes scene 6 alone.

💻 Load the program

GROWN-UP JOB

A grown-up loads the program (main.ts) from the website: **https://pandagardenio.github.io/robokids/web/phase-5-robots-that-learn/week-47-ai-story-robot/**. Each week's page has a big Copy button and the steps.

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
AI	Artificial intelligence: a computer program that learns from examples instead of following rules someone wrote.	A computer that learns by looking at examples.
AI assistant	A chat tool like Claude or ChatGPT. At this age, the parent types and reads.	A computer helper that writes words.
List	A set of items in order. The program goes through them one by one.	Like a shopping list for the robot.
Code	Another word for the program: the instructions the robot follows.	The robot's instructions.

Week 47 · AI Story Robot

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

48

PHASE 5 · ROBOTS THAT LEARN

Talking Robot

~45 min build + short play sessions during the week · Skill: Speech output

Done! ☐

🧰 You need

- ☐ Robot car or micro:bit by USB
- ☐ Laptop with speakers and the AI bridge

🔧 Build together

TOGETHER

1. Load `main.ts`. Close MakeCode, open the AI bridge, choose **No model (talking robot only)** and click **Connect micro:bit**.
2. The robot sends `say:` lines to the laptop, which speaks them.

🎮 Play

KID TIME

Interview the robot: A = 'What's your name?', B = 'Tell a joke', shake = 'What's your favorite food?'

💬 Talk about it

TOGETHER

Is the robot really talking, or is the laptop helping?

⭐ Level up

KID TIME

Kid writes the robot's answers (you type them in).

💻 Load the program

GROWN-UP JOB

A grown-up loads the program (main.ts) from the website: **https://pandagardenio.github.io/robokids/web/phase-5-robots-that-learn/week-48-talking-robot/**. Each week's page has a big Copy button and the steps.

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
Speech	The laptop reads text out loud with a computer voice.	The laptop reads the robot's words out loud.
Output	Something the robot does: lights, sounds, moving.	What the robot does.
USB	The cable between the laptop and the micro:bit. It carries messages, not just power.	The cable that lets the laptop talk to the robot.
AI	Artificial intelligence: a computer program that learns from examples instead of following rules someone wrote.	A computer that learns by looking at examples.
AI bridge	A page from this project that passes what the AI sees or hears to the micro:bit. See the AI bridge guide.	The messenger between the AI and the robot.
Model	What the AI ends up with after training: its 'memory' of what it learned. Exporting it gives you a link.	The AI's memory.
Accelerometer	The sensor that feels movement: shaking, tilting, falling.	It feels when you shake or tip the robot.

Week 48 · Talking Robot

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

PHASE 5 · ROBOTS THAT LEARN

49 Dream Robot Build

~45 min build + short play sessions during the week · Skill: Design thinking

Done! ☐

🧰 You need

- ☐ Everything from the year
- ☐ Cardboard, tape, Robot Journal

🔧 Build together

TOGETHER

1. Kid draws the dream robot: what it looks like, what it does, which sensors it uses.
2. Build the cardboard body around the car.
3. Start from `main.ts`: a template with slots for button, shake, distance and sound behaviors.

🎮 Play

KID TIME

Rehearse the robot's show for next week.

💬 Talk about it

TOGETHER

What was easy to build, and what needed a change from the drawing?

⭐ Level up

KID TIME

Add one AI behavior from weeks 42–48.

💻 Load the program

GROWN-UP JOB

A grown-up loads the program (main.ts) from the website: **https://pandagardenio.github.io/robokids/web/phase-5-robots-that-learn/week-49-dream-robot/**. Each week's page has a big Copy button and the steps.

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
Sensor	A part that lets the robot notice the world: light, heat, distance, sound, movement.	The robot's eyes, ears and skin.
Template	A ready-made program with empty slots to fill in.	A robot program with blanks to fill.
Accelerometer	The sensor that feels movement: shaking, tilting, falling.	It feels when you shake or tip the robot.
Behavior	One thing the robot does in response to something, like 'back away when it's loud'.	Something the robot does when something happens.
AI	Artificial intelligence: a computer program that learns from examples instead of following rules someone wrote.	A computer that learns by looking at examples.

Week 49 · Dream Robot Build

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

PHASE 5 · ROBOTS THAT LEARN

50 Grand Finale: Robot Expo

~45 min build + short play sessions during the week · Skill: Celebrate the year

Done! ☐

🧰 You need

- ☐ Favorite robots from all phases
- ☐ Certificate (see `certificate.md`)
- ☐ Family audience

🔧 Build together

TOGETHER

1. Set up three stations: a Phase 1 robot, a micro:bit project, and the dream robot.
2. Kid prepares one sentence per robot.

🎮 Play

KID TIME

Robot Expo! Kid presents, family asks questions. Hand over the certificate.

💬 Talk about it

TOGETHER

What do you want to build next year?

⭐ Level up

KID TIME

Pick a project to keep improving in year 2.

Week 50 · Grand Finale: Robot Expo

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____

BONUS PROJECT · FITS AFTER WEEK 37

Robot Mouse 🐭

2–3 sessions · Skill: touch sensors, behaviors, radio signal strength

Done! ☐

🔀 Pick your version

	Cutebot mouse	DIY mouse
Difficulty	Easy: a costume on the car you have	Medium: you build the chassis
Time	1 afternoon	2–3 weekends
Whiskers	Ultrasonic sensor (feels things before touching)	Real paper-clip whiskers
Size	~10 cm	~12 cm (rat-sized is easier than mouse-sized)
Extra cost	~0 €	~25 €
Program	`main.ts`	`diy-mouse.ts`

Both use `cheese.ts` on a second micro:bit as the hidden cheese.

🧰 You need

Both versions - micro:bit V2 (for the squeak and clap detection) - Second micro:bit + battery pack for the cheese - Grey or brown felt, a paper cup or yogurt pot for the body, a pipe cleaner or string tail, felt ears, two googly eyes - Yellow card for a cardboard cheese wedge

Cutebot mouse: your Cutebot. Optional: the SG90 servo from week 31 on S1 as a wagging tail.

DIY mouse - Kitronik Motor Driver Board for micro:bit (or similar) - 2 N20 micro gear motors (~100–300 RPM, 3–6 V) with 30–40 mm wheels - 1 small ball caster - 3×AAA battery holder with switch - 2 large paper clips, 3 jumper wires with crocodile clips, a small piece of stiff card for the base - Hot glue (parent)

🔧 Build: Cutebot mouse

TOGETHER

1. Cut a paper cup in half lengthwise and cover it in felt: that's the body.
2. Glue on ears and googly eyes at the front, above the ultrasonic sensor (its two "eyes" stay uncovered).
3. Tape the shell over the car, leaving the micro:bit screen visible as the mouse's "heart window".
4. Optional tail: tape a pipe cleaner to the servo horn on S1.
5. Load `main.ts` with the **Cutebot** extension (see Load the programs).

🔧 Build: DIY mouse

TOGETHER

1. **Base:** cut a 7 × 11 cm card base, rounded at the front. Hot glue the two motors at the back, one on each side, shafts pointing out. Push the wheels on.
2. **Balance:** glue the ball caster under the front. The mouse should sit level.
3. **Electronics:** plug the micro:bit into the motor board and glue the board on top. Wire the left motor to Motor 1 and the right motor to Motor 2. Battery pack under or behind the board.
4. **Whiskers:** bend each paper clip into a long, springy whisker and fix one to each front corner. Connect the left whisker to **P1**, the right to **P2**. Run a bare GND wire across the front, just behind both whiskers, so a bent whisker touches it. (Check your board's labels for where P1, P2 and GND are broken out; crocodile clips on the micro:bit's edge pads also work.)
5. **Test before decorating:** load `diy-mouse.ts` with the **Kitronik Motor Driver** extension (see Load the programs). If the mouse spins instead of driving forward, swap one motor's two wires.
6. **Costume:** felt body over the top, ears, eyes, tail. Keep the whiskers free to move.

🧀 Cheese

Load `cheese.ts` onto the second micro:bit and hide it inside the cardboard cheese wedge.

💻 Load the programs

GROWN-UP JOB

A grown-up loads the program (main.ts, cheese.ts, diy-mouse.ts) from the website: **https://pandagardenio.github.io/robokids/web/bonus/robot-mouse/**. Each week's page has a big Copy button and the steps.

🎮 Play

KID TIME

Do this	The mouse…
Leave it alone	Scurries, stops to sniff, scurries again
Whisker touches a wall	Squeaks, backs up and turns away
Clap loudly	Runs away and spins
Turn off the lights	Freezes and falls asleep
Press B	Hunts for the cheese. When it finds it: heart, tail wag, double squeak
Press A	Stops hunting and goes back to wandering

Games: hide the cheese in a cardboard maze and time the hunt. Play "cat": your kid claps when the mouse gets too close to the cat's corner. Build a mouse hole from a shoebox and see if it finds its way home.

💬 Talk about it

TOGETHER

How does the mouse know it touched something? How do real mice do it?

Real mice use their whiskers to feel their way in the dark, just like this robot.

⭐ Level up

KID TIME

- Make the mouse squeak louder (more notes) the closer it gets to the cheese.
- Add a "hungry" counter that grows over time: the hungrier it is, the faster it scurries.
- Build a second mouse and have them chase each other by radio, like week 39.
- Parent challenge: turn it into a wall-following maze mouse, the first step towards Micromouse.

🎛️ Tuning knobs

TOGETHER

- `whiskerCm` (Cutebot): how close something must be before the mouse "feels" it.
- `SoundThreshold.Loud, 160`: raise it if the mouse runs away from normal talking.
- `s > -48`: how close counts as "found the cheese". Lower it (e.g. `-55`) if it never finds it.
- `lightLevel() < 15`: how dark before it sleeps.

📖 Words this week

FOR GROWN-UPS

Word	What it means	Say it to your kid
Sensor	A part that lets the robot notice the world: light, heat, distance, sound, movement.	The robot's eyes, ears and skin.
Behavior	One thing the robot does in response to something, like 'back away when it's loud'.	Something the robot does when something happens.
Radio	micro:bits can send short messages to each other through the air, no wires needed, across a house.	Invisible messages through the air.
Signal strength	How strong a radio message is when it arrives. Closer usually means stronger.	Louder when close, quieter when far.
Whisker sensor	A springy wire that touches another wire when it bends, closing a circuit the micro:bit can feel. A homemade touch sensor.	When the whisker bumps something, the robot feels it.
Cutebot	The robot car kit this project uses. The micro:bit slots in as its brain.	Our robot car.
Chassis	The base of a vehicle that everything else is attached to.	The robot's skeleton.
Ultrasonic sensor	The two round 'eyes' on the front of the Cutebot. They send out a sound too high to hear and time the echo to measure distance.	The robot squeaks and listens for the echo, like a bat.
V2	The newer micro:bit, with a speaker, a microphone and a touch-sensitive logo. This project needs it.	The newer micro:bit that can hear and talk.
Servo	A motor that turns to an exact angle and holds still there, instead of spinning round and round.	A motor that points instead of spins.
S1	A socket on the Cutebot labeled S1 where you plug in a servo.	The servo's plug spot.
Motor driver board	A board that gives motors enough power, with the micro:bit telling it what to do.	The muscles helper for the motors.
Gear motor	A motor with gears inside: slower but stronger, good for turning wheels.	A slow, strong motor for wheels.
RPM	Revolutions per minute: how many full turns a motor makes in one minute.	How fast a wheel spins.
Volts (V)	How hard a battery pushes electricity. One AA battery is 1.5 V; two make 3 V.	How strong the battery's push is.
Ball caster	A small ball in a holder that lets the front of a robot slide in any direction.	A rolling ball foot.
Switch	Something that opens or closes a gap in a circuit, to turn things off and on.	A little door for electricity: open means off, closed means on.
Jumper wire	A short wire for connecting parts, often with clips or pins on the ends.	A short connecting wire.
Alligator clip	A wire with a spring clip at each end, for connecting parts without tools.	A wire with crocodile teeth.
Servo horn	The small plastic arm that clips onto the servo and moves with it.	The servo's little arm.
Shaft	The metal rod that sticks out of a motor and spins.	The motor's spinning nose.
Balance	How weight is spread out. Low and heavy robots are harder to tip over.	Low and heavy is hard to push over.
Pins (P1, P2, GND)	The metal strips along the micro:bit's bottom edge. P1 and P2 can feel when they're connected to GND (the minus side).	The micro:bit's fingers along the bottom.

Bonus · Robot Mouse 🐭

✏️ My Robot Journal

Draw your robot here!

My robot is called _____

It can _____

_____

How much fun? ☆ ☆ ☆ ☆ ☆

Date _____